Unit - V

Statistical Learning and Applications

Linear Regression

Linear regression is a supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous numerical values by fitting a straight line that best represents the data.

Linear Regression

$$\hat{Y} = a + bX$$

Linear Regression

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Predicted Value (Dependent)

Linear Regression

$$\hat{Y} = a + bX$$

Intercept (Value of Y when X = 0)

Linear Regression

$$\hat{Y} = a + bX$$

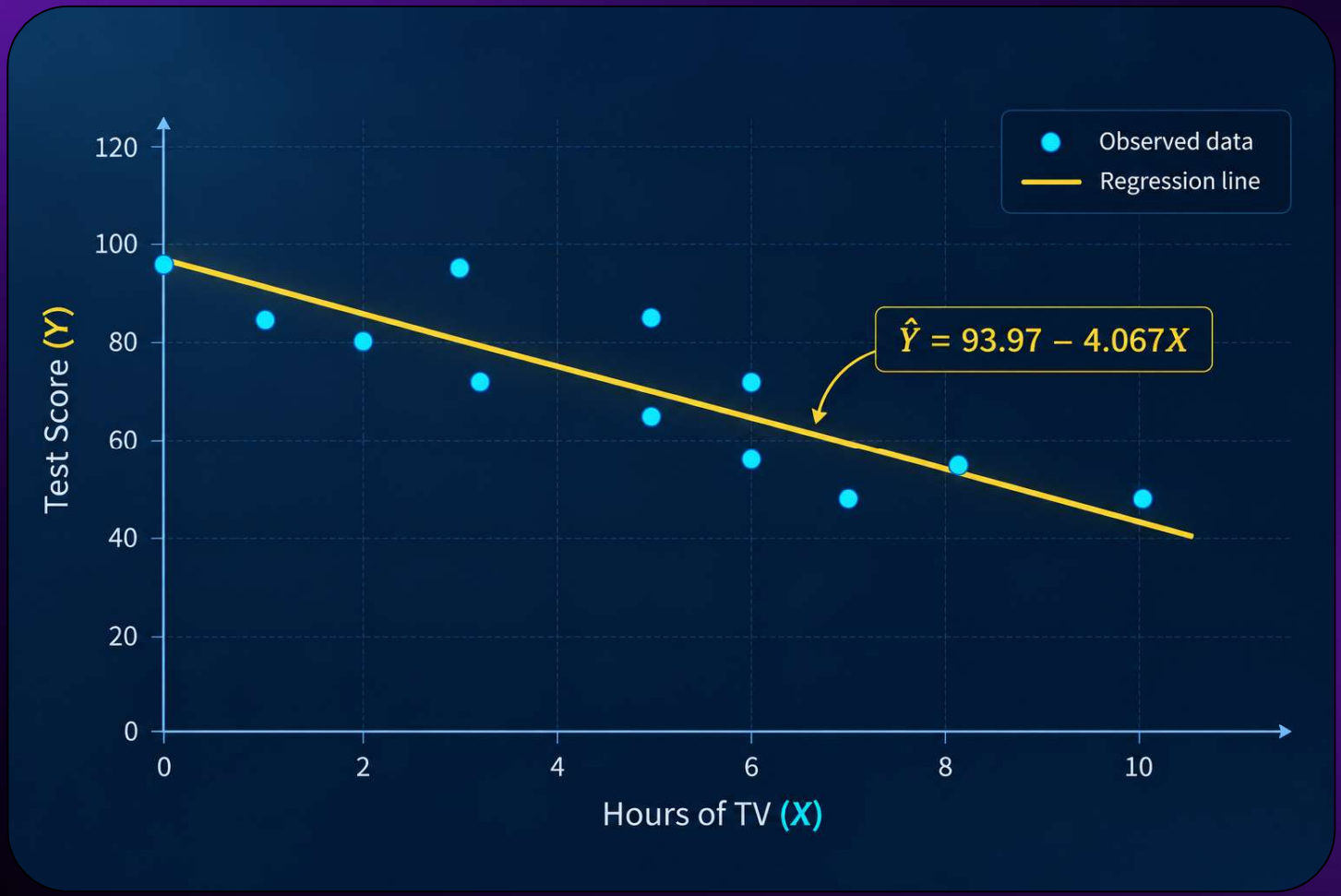
Slope (Change in Y for a 1 – unit change in X)

Linear Regression

$$\hat{Y} = a + bX$$

Independent Variable

Linear Regression



Linear Regression

Example 1: The following data represents the number of hours, 12 different students watched television during the weekend and the scores of each student who took a test the following Monday.

a) Find the equation of the regression line.
b) Use the equation to find the expected test score for a student who watches 9 hours of TV.

Hours, X	0	1	2	3	3	5	5	5	6	7	7	10
Test Score, Y	96	85	82	74	95	68	76	84	58	65	75	50

Linear Regression

Hours (X)	Test Score (Y)
0	96
1	85
2	82
3	74
3	95
5	68
5	76
5	84
6	58
7	65
7	75
10	50

Linear Regression

Hours (X)	Test Score (Y)	XY	x^2
0	96	0	0
1	85	85	1
2	82	164	4
3	74	222	9
3	95	285	9
5	68	340	25
5	76	380	25
5	84	420	25
6	58	348	36
7	65	455	49
7	75	525	49
10	50	500	100
$\Sigma X = 54$	$\Sigma Y = 908$	$\Sigma XY = 3724$	$\Sigma x^2 = 332$

Linear Regression

Hours (X)	Test Score (Y)	XY	x^2
0	96	0	0
1	85	85	1
2	82	164	4
3	74	222	9
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The Normal equations of Y = a + b X

$$\sum Y_i = n\,a + b \sum X_i \quad \rightarrow (1)$$

$$\sum X_i Y_i = a\,X_i + b \sum X_i^2 \quad \rightarrow (2)$$

Linear Regression

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Substitute ΣX, ΣY, ΣXY Σx^2 values in the normal equations, then we get

$908 = 12\,a + 54\,b \rightarrow (3)$

$3724 = 54\,a + 332\,b \rightarrow (4)$

Linear Regression

Solve eq (3) &eq (4) for the values of a & b

$$908 = 12\,a + 54\,b \rightarrow (3)$$

$$3724 = 54\,a + 332\,b \rightarrow (4)$$

Linear Regression

We get,

$a = 93.97 \qquad b = -4.067$

Linear Regression

Then the predicted linear regression line is

$$\hat{Y} = 93.97 - 4.067\,(X)$$

Linear Regression

a.) Equation of the regression line :

$$\hat{Y} = 93.97 - 4.067\,(X)$$

Using the equation we can predict the test score for a student who watches 9 hours of TV.

Linear Regression

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$$\hat{Y} = 93.97 - 4.067\,(9)$$

Linear Regression

a.) Equation of the regression line :

$$\hat{Y} = 93.97 - 4.067\,(X)$$

Using the equation we can predict the test score for a student who watches 9 hours of TV.

$$= 57.34$$

Linear Regression

a.) Equation of the regression line :

$$\hat{Y} = 93.97 - 4.067\,(X)$$

b.) Expected test score for a student who watches 9 hours of TV :

$$57.34$$